

- 5 (a) The Principle of Superposition states that when two or more waves meet at a point, the resultant displacement at that point is equal to the vector sum of the displacements of the individual waves at that point [B1].

(b) (i)
$$x = \frac{\lambda D}{a} = \frac{(590 \times 10^{-9})(3.2)}{1.50 \times 10^{-3}} \quad [\text{C1}]$$

$$x = 0.00126 \text{ m} \quad [\text{A1}]$$

- (ii) As the slits are widened, the extent of diffraction of the waves when they passes through the slits will become lesser. [M1]
 This will result in the waves not being able to overlap one another resulting in no interference[M1] and thus no fringes formed [A0]

(iii)
$$\tan \theta = \frac{1.5}{3.2}$$

$$\theta = 25.1^\circ \quad [\text{C1}]$$

$$d \sin \theta = n\lambda$$

$$n = \frac{d \sin \theta}{\lambda} = \frac{0.001 \sin(25.1)}{590 \times 10^{-9}} \quad [\text{C1}]$$

$$n = 1.03 = 1.0 \quad [\text{C1}]$$

 Max. bright fringes = $n \times 2 + 1 = 3 \quad [\text{A1}]$

[1] for finding d
 [1] diffraction formula

